Class 06 Homework: Q6 Protein B-factor Function

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Q6. Generalizing the original code

The original code does the same four steps for every protein:

1. read the protein with `read.pdb()`
2. keep only chain A alpha carbons with `trim.pdb()`
3. get the B-factor values from `$atom$b`
4. plot the B-factors with `plotb3()`

Because these steps are copied for each protein, it is easy to make copy/paste mistakes. In the original code, `s3.chainA` was made from `s1` instead of `s3`. To fix this, I wrote the steps once inside a function. The PDB code is the input, so the function works for any protein.

Inputs

The function has one input:

- `pdb.id`: the 4 letter PDB code of the protein, written in quotes (for example `"4AKE"`).

What the function does and how to use it

The function reads in the protein structure, trims it down to chain A alpha carbons (one atom per amino acid), pulls out the B-factor values, and plots them. The plot also exhibits the secondary structure being the following: black rectangles are alpha helices and grey rectangles are beta strands.

To use it, load the bio3d package and call the function with a PDB code:

```
plot_protein("4AKE")
```

Output

The function draws a B-factor plot for the protein, with the PDB code as the title. Furthermore, it returns the B-factor values as a vector, so they can be saved and used later like comparing proteins.

Function code

```
library(bio3d)

plot_protein <- function(pdb.id) {
  # read in the protein structure
  s <- read.pdb(pdb.id)

  # keep only chain A alpha carbons
  s.chainA <- trim.pdb(s, chain="A", elety="CA")
```

```
  # get the B-factor values
  s.b <- s.chainA$atom$b

  # plot the B-factors with secondary structure in the margins
  plotb3(s.b, sse=s.chainA, typ="l", ylab="Bfactor", main=pdb.id)

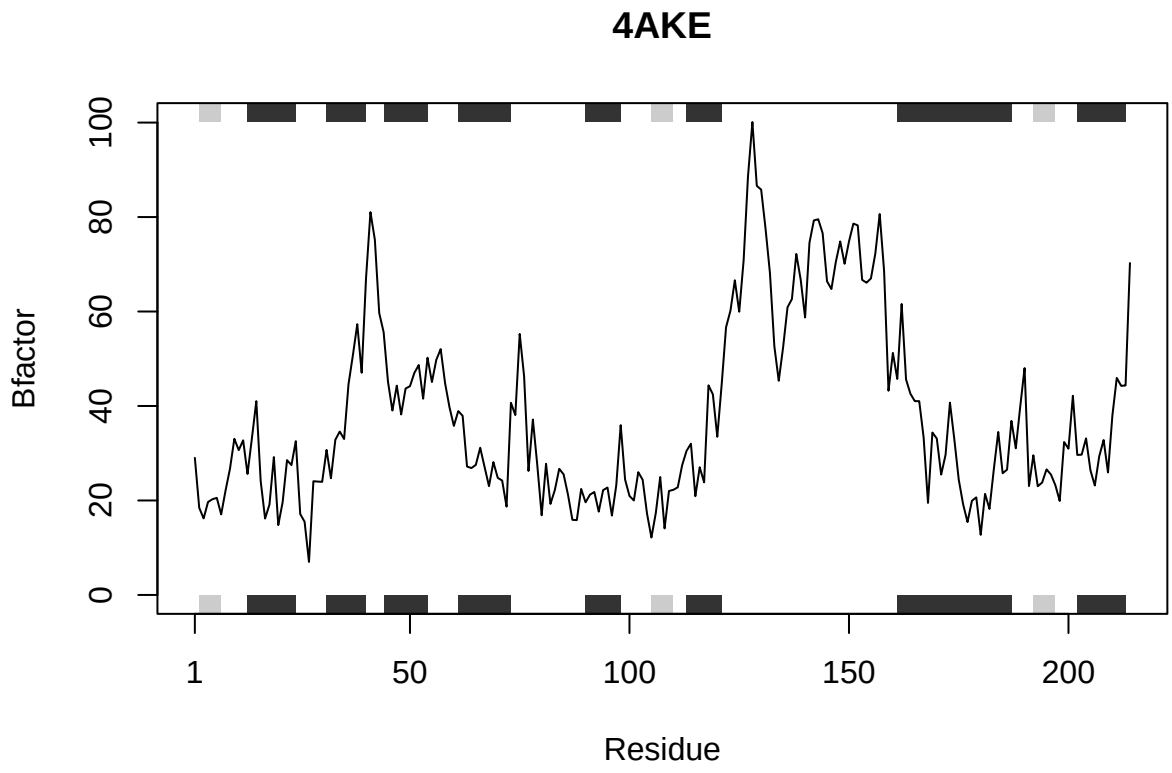
  # return the B-factor values
  return(s.b)
}
```

Example output

Calling the function on the three proteins from the original code:

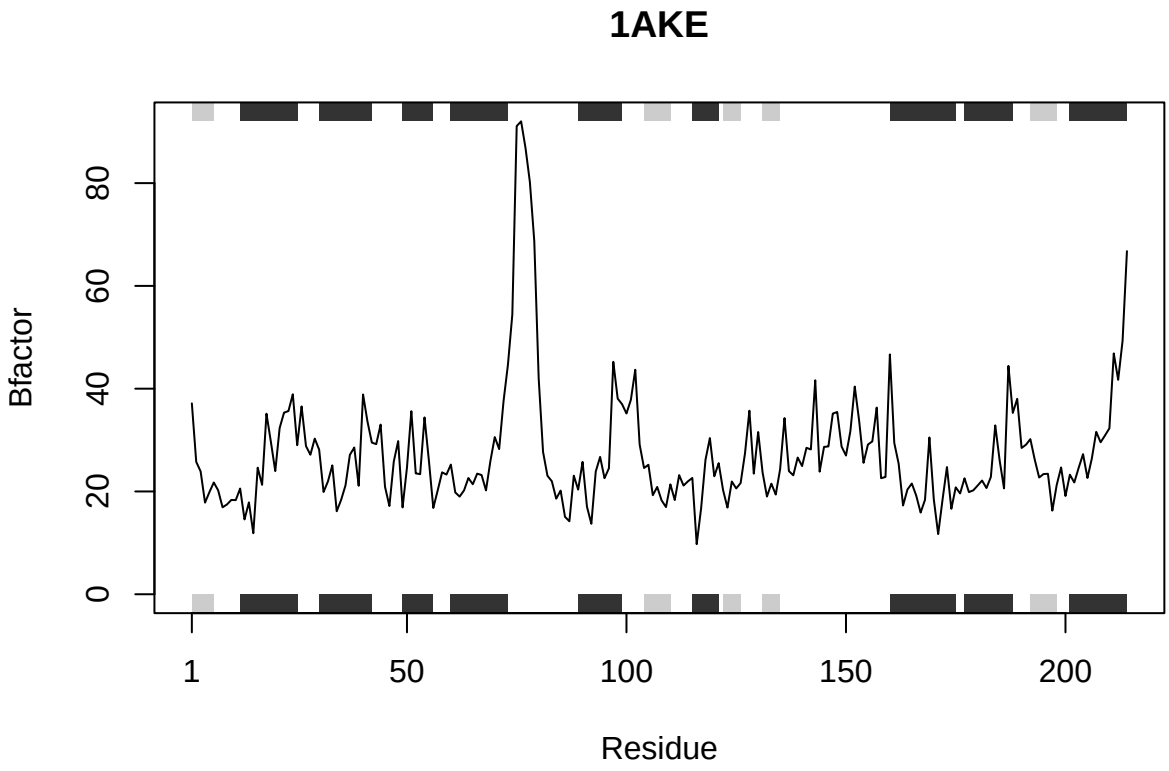
```
s1.b <- plot_protein("4AKE")  # kinase with drug
```

```
##   Note: Accessing on-line PDB file
```



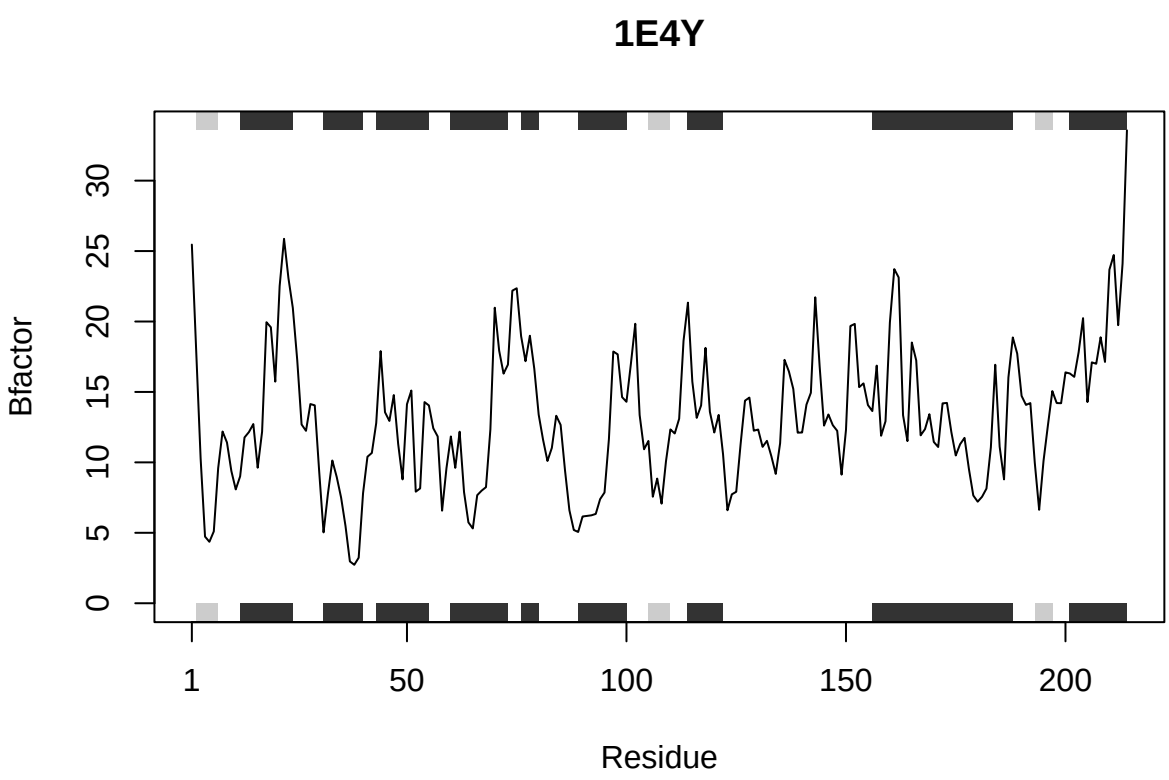

```
s2.b <- plot_protein("1AKE")  # kinase no drug
```

```
##   Note: Accessing on-line PDB file
##    PDB has ALT records, taking A only, rm.alt=TRUE
```




```
s3.b <- plot_protein("1E4Y")  # kinase with drug
```

```
##   Note: Accessing on-line PDB file
```



Why this is better than the original code

- **Less repeated code:** steps are written once instead of three times.
- **No copy/paste errors:** the s1/s3 mistake from the original can't happen.
- **Works for any protein:** analyzing a new protein now takes only one line, like `plot_protein("1HSG")`.